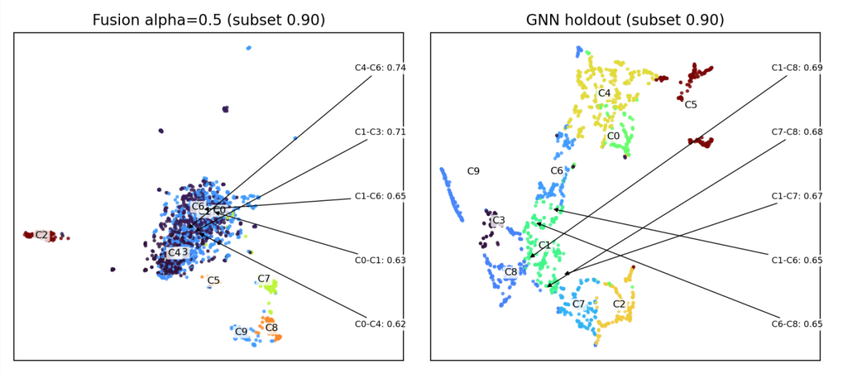
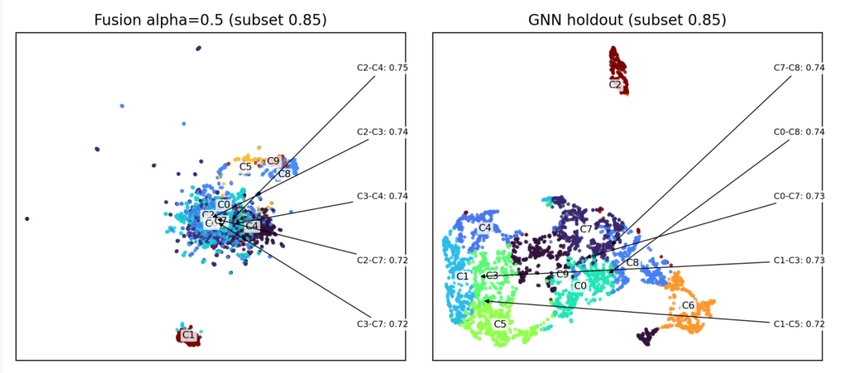
Alpha=0.5 Fusion vs GNN Holdout: Poster Summary (Subset 0.85 + 0.90)

Data source: task_20260217_metrics_080_085 + task_20260217_alpha05_subset090 | k=10 primary comparison

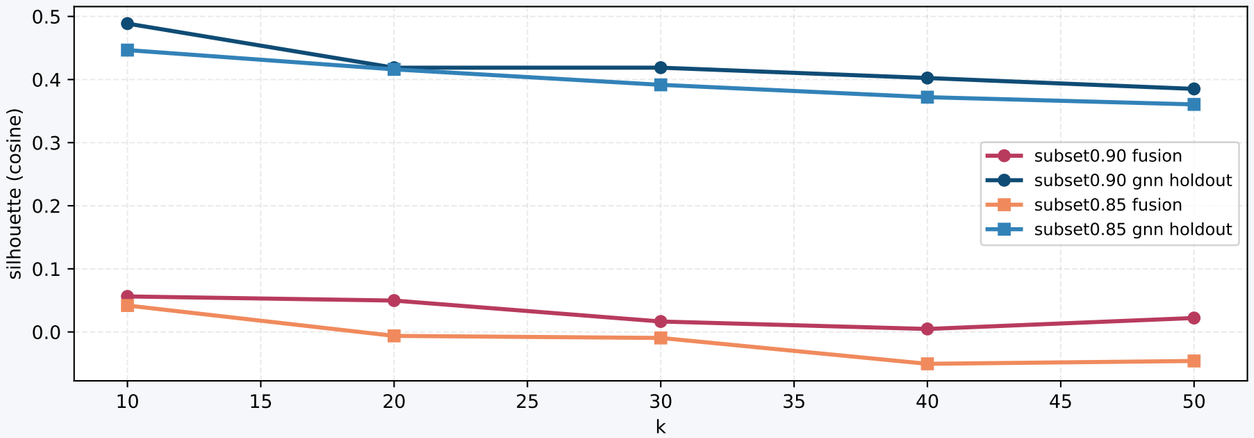
Subset 0.90 | UMAP + super-PAG overlap



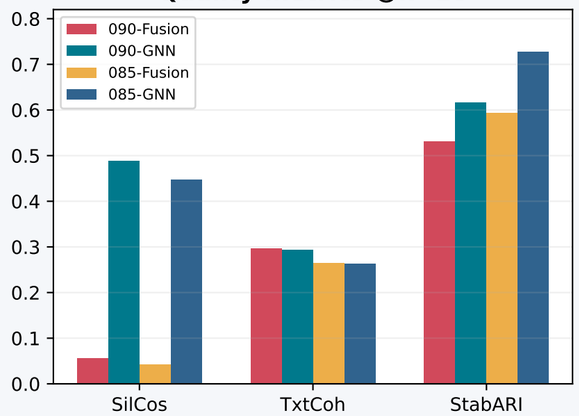
Subset 0.85 | UMAP + super-PAG overlap



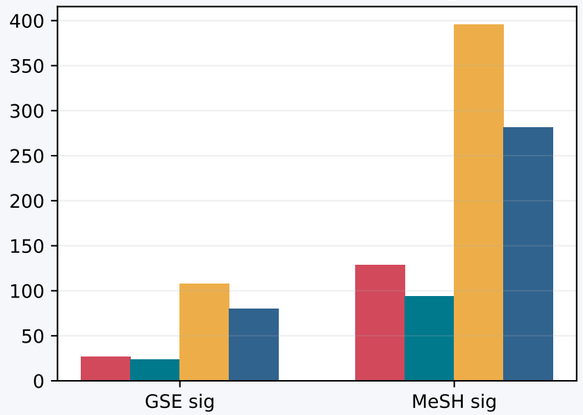
Cosine silhouette vs k



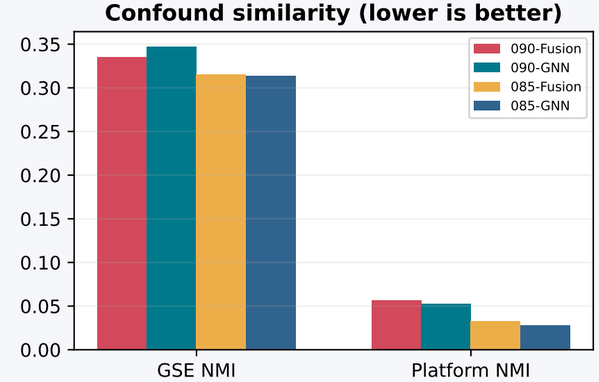
Quality metrics @k=10



Meta enrichment counts @k=10



Confound similarity (lower is better)



Key takeaways

1) Internal separation: GNN dominates on silhouette_cosine @k=10 (subset0.90: 0.489 vs fusion 0.056; subset0.85: 0.447 vs fusion 0.042).
2) Text coherence: fusion slightly higher (subset0.90 clean=0.296 vs gnn=0.293; subset0.85 clean=0.264 vs gnn=0.262).
3) Stability: GNN pairwise ARI is higher (subset0.90 0.616 vs fusion 0.531; subset0.85 0.727 vs fusion 0.594).
4) Meta enrichment: fusion has larger significant-hit counts for GSE/MeSH in both subsets (e.g., subset0.85 MeSH 395.8 vs gnn 281.0).
5) Confound metrics are close: primary_gse_nmi (subset0.90 fusion 0.335, gnn 0.347); no up/down conflict records detected (both CSVs contain only header).